

Math 142, S12

Quiz 12

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (6 points) The Cartesian coordinates of a point are given.

(a) $(-1, \sqrt{3})$. Find the polar coordinates (r, θ) of the point where $r > 0$ and $0 \leq \theta < 2\pi$.

$$r^2 = x^2 + y^2 = (-1)^2 + (\sqrt{3})^2 = 4 \Rightarrow r = 2. \quad (\text{Since } r > 0).$$

$$\tan \theta = \frac{y}{x} = \frac{\sqrt{3}}{-1} = -\sqrt{3}$$

$(-1, \sqrt{3})$ is in the second quadrant $\Rightarrow \theta = \frac{2\pi}{3}$

$\Rightarrow$ The polar coordinates $(2, \frac{2\pi}{3})$.

(b) $(3, -3)$. Find the polar coordinates (r, θ) of the point where $r < 0$ and $0 \leq \theta < 2\pi$.

$$r^2 = 3^2 + (-3)^2 = 18 \Rightarrow r = 3\sqrt{2} \text{ or } -3\sqrt{2}$$

$$\tan \theta = \frac{-3}{3} = -1$$

$(3, -3)$ is in the fourth quadrant $\Rightarrow \theta = \frac{7\pi}{4}$

Since it asks $r < 0$, we have the polar coordinates

$$(-3\sqrt{2}, \frac{7\pi}{4} - \pi) = (-3\sqrt{2}, \frac{3\pi}{4})$$

Problem 2. (4 points) Identify the curve by finding a Cartesian equation for the curve:

$$r = 3 \sin \theta.$$

Solution: $r = 3 \sin \theta \Rightarrow r^2 = 3 r \sin \theta.$

$$x^2 + y^2 = 3y$$

$$x^2 + y^2 - 3y = 0$$

$$x^2 + (y - \frac{3}{2})^2 = (\frac{3}{2})^2.$$

$\Rightarrow$ A circle centered at $(0, \frac{3}{2})$ with radius $\frac{3}{2}$.